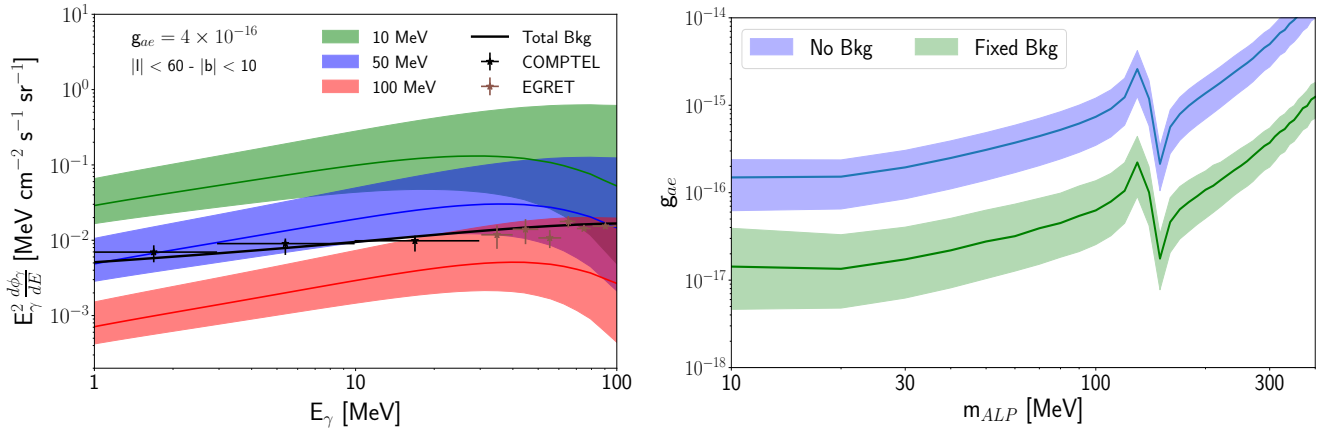

FIG. 7. Effect of propagation uncertainties on the predicted IA signals from ALPs produced in SNe (left panel) for a few different masses. For completeness, we also show here our reference background model (model prediction from Ref. [118]) and COMPTEL and EGRET measurements. The right panel shows the effect of propagation uncertainties on the constraints (at 2σ confidence level) on the ALP coupling, in the conservative case, in which we do not include any background emission ("No Bkg"), and in the case in which we account for background when deriving these limits labeled as "Fixed Bkg" in the legend of the figure.

model the background as a simple power-law and get the constraint comparing the sum of the background plus the IA FIP signals. In a second case, we make use of a more refined background model. In particular, we use the electron model from Refs. [118, 119], which is optimized to reproduce the electron and positron emission at Earth location, as well as the local γ ray emissivity, down to a few tens of MeV. We have checked that this model of the γ-ray background emission shows a notable agreement with data down to 100 keV, for different regions of interest. We will refer to this model as our reference model. Using this reference model, which is shown as a black line in Fig. 5, we compute constraints not only for the case of a fixed background model, but also leaving the normalization of the background emission free in our analysis. Therefore, in addition to the conservative limits without any background, we obtain three other constraints: one assuming a power-law background, another one from the predicted background model explained above and a last one letting the normalization of this background model be free.

To perform these fits and derive the constraints we rely on the Markov Chain Monte Carlo (MCMC) package *Emcee* [126], that is based on Bayesian inference, since this technique is more robust than conventional optimizers and less prone to finding false local minima. This analysis provides the probability distribution functions for every parameter to reproduce the data, and, thus, we obtain, as an output, the credible (confidence) intervals of each parameter included in the fit (normalization of the background, spectral index, coupling of the FIP, etc.). Having the credible intervals for each parameter, we obtain the limits at the 95% confidence level on the FIP coupling. A similar setup has been successfully used in

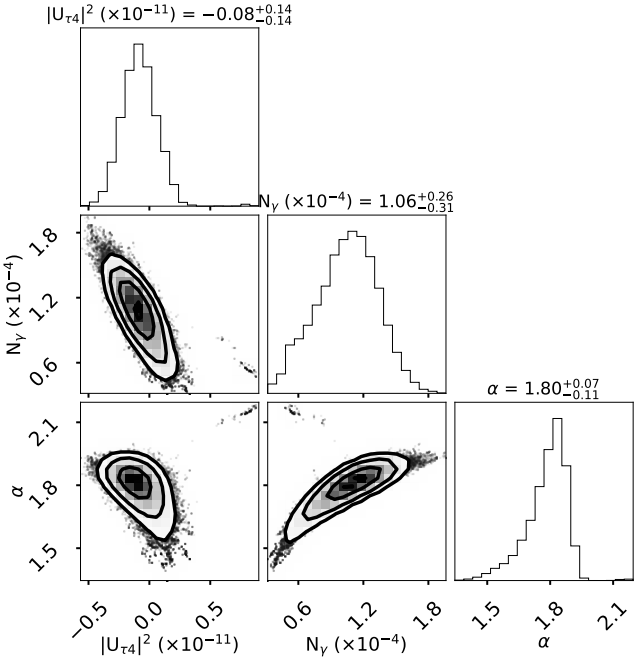

FIG. 8. Example of the probability distribution and credible intervals for the parameters included in the analysis of an IA signal from a 30 MeV sterile neutrino when adding the background contribution described as a power-law (with free parameters the normalization at 10 MeV, "N_γ", in units of MeV cm^{-2} s^{-1} sr^{-1}, and spectral index, "α"). On top of each panel, the best-fit values and their 1σ uncertainty are shown.

different studies involving dark matter limits from antiprotons [127, 128], line searches in γ-ray data [129] or even to study propagation of CRs [103].

Firstly, we investigated if there is any evidence for IA